

Multiple Choice (10 marks)

1. Cells take in food for energy. The part of the cell that aids in digestion of the food is the lysosome. What is the main role of lysosomes in the process of food digestion?
 - (a) building proteins
 - (b) breaking down wastes
 - (c) controlling the activities of the cell
 - (d) converting energy from one form into another
2. Which of these activities will result in a different kind of material being formed?
 - (a) A nail is left outside and it rusts.
 - (b) A glass is dropped and it shatters into small pieces.
 - (c) A rubber band is stretched until it breaks.
 - (d) A pencil is sharpened to a point.
3. The same side of the Moon is always visible from Earth because
 - (a) the Moon does not rotate on its axis.
 - (b) the far side of the Moon faces Earth during the day.
 - (c) the Moon makes one rotation per revolution around Earth.
 - (d) the rate of rotation is the same for Earth and the Moon.
4. A statue and a table are both made of the same type of marble. Which of the following properties will most likely be the same for both of these objects?
 - (a) size
 - (b) shape
 - (c) weight
 - (d) hardness
5. Which statement best describes a structural adaptation that helps an organism survive in a desert?
 - (a) A bird builds its nest in a cactus.
 - (b) A rattlesnake shakes its tail when threatened.
 - (c) A lizard crawls under a rock for shelter.
 - (d) The ears of a jackrabbit release heat.

True / False (10 marks)

Answer True or False

6. True or False: Heat energy from the Sun is transferred to Earth primarily through convection.
7. True or False: A standardized taxonomic classification system is primarily used to determine the geographical habitats of all organisms.
8. True or False: The average distance between Earth and the Sun is 149.6×10^6 kilometers, which is not in proper scientific notation.
9. True or False: When a battery-operated flashlight is turned on, electrical energy is transformed into light energy.
10. True or False: Heating kernels on different types of stoves is sufficient to find out which popcorn type produces the most popped kernels.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the processes involved in the formation of a beach surface, including the role of natural elements such as water, wind, and sediment.
12. How do earthquakes contribute to our understanding of Earth's geological history, and what can they reveal about past tectonic activity and environmental changes?

End of Paper